Aryan Roy

aryan.roy4400@gmail.com | 943-235-2172 | Website | Google Scholar

EDUCATION

Georgia Institute of Technology — Atlanta, GA
Master of Science in Computer Science : Machine Learning Specilization — GPA: N/A — Exp: May, 2027

Manipal Institute of Technology — India
Bachelor of Technology in Computer and Communication Engineering — GPA: 8.52/10 — July, 2023

WORK EXPERIENCE

Oracle — India
Software Engineer, Reporting and Analytics Team — July, 2023 – August, 2025

- Delivered "Ask Oracle", an intelligent multi-data source Q&A assistant for enterprise customers.
- Developed an ML based sales forecasting engine as proof of concept with an average error rate of 4.2%.
- Developed and integrated a monitoring service for OCI Streaming + Kafka, leveraging Prometheus and OCI monitoring tools.
- Promoted to IC2 after one year at Oracle.

Oracle — India
Project Intern, Reporting and Analytics Team — January, 2023 – July, 2023

- Engineered benchmark scripting suite using Python and Bash to evaluate web application load response times under varying simulated user traffic conditions.
- Implemented an API activity monitoring service to provide insights into STS API traffic with $>$ 1 million hits/day.

Fermi National Accelerator Lab — Batavia, IL
Research Intern, Computing Division — May, 2022 – July, 2022

- Integrated a lightweight experimental DSL into a widely used Python I/O library for runtime parsing of serialized objects under the mentorship of Dr. Lindsey Gray.
- Achieved a 400x gain in performance when reading data types of interest.

Princeton University — Remote
Research Intern — April, 2021 – August, 2021

- Developed a Python library to perform sequential recombination clustering for particle physics data under the guidance of Dr. Jim Pivarski.
- Accelerated throughput by 25x compared to alternatives.

RESEARCH AND OPEN SOURCE CONTRIBUTIONS

Graduate Researcher *Teachable AI Lab, Georgia Tech* — August, 2025 – Present

- Investigating human factors in AI based tools/application and internal world representation in generative models under Dr. Christopher MacLellan.

Open-Source Contributor *pywhy-graphs* — January, 2023 – May, 2024

- Implemented fundamental theorems and functionalities in statistical causality under the mentorship of Dr. Adam Li.

Open-Source Contributor *Uproot* — January, 2023 – Dec, 2023

- Designed and implemented a Python based transpiler using Lark to translate between 3 standard scientific notations in High Energy Physics under Dr. Jim Pivarski.

ACTIVITIES

Graduate Teaching Assistant *CS 4635 A/7637* — Fall Semester, 2025

- Awarded a competitive teaching fellowship with tuition waiver.
- Supported instruction for Knowledge-Based AI (CS 4635/7637) by designing and grading assignments and quizzes.
- Held office hours and mentored students, addressing coursework and conceptual queries.

AWARDS

IRIS-HEP Fellowship (2021,2022): Awarded the Fellowship twice to work on implementing solutions for HPC workflows in High Energy Physics.

MITACS GRI (2022): Awarded the fully funded scholarship to work on neurologically inspired authentication tools.

SKILLS

Programming Languages: Python, Java, Forth
Technologies: JAX-RS, Pytorch, scikit-learn, Kafka, CI/CD, Agentic AI
Tools: Prometheus, Github Actions, Conda
Soft Skills: Public Speaking, Technical Writing